

End-to-End Amazon S3 Security Implementation

Cloud Security Assessment & Implementation Report

Prepared By

Nitin Sukthe

Role

Cloud Security Engineer (Portfolio Project)

Date

June 2026

Executive Summary

This project focused on securing data stored within Amazon Simple Storage Service (Amazon S3) through the implementation of multiple cloud security controls aligned with AWS security best practices.

The objective was to design and configure a secure storage environment capable of protecting sensitive information from unauthorized access, accidental exposure, data loss, and insecure communication channels.

The implementation included public access restrictions, encryption using AWS Key Management Service (AWS KMS), bucket policy enforcement, object versioning, access logging, and recovery validation.

The project successfully demonstrated a defense-in-depth approach to cloud storage security and improved the overall security posture of the Amazon S3 environment.

Project Objectives

The primary objectives of this project were:

- Create a secure Amazon S3 storage environment
- Prevent public exposure of cloud-hosted data
- Implement encryption for data-at-rest protection
- Enforce secure communication using HTTPS
- Enable audit logging and monitoring
- Configure versioning for recovery and resilience

- Demonstrate cloud security best practices
-

Scope

Included Components

- Amazon S3
- AWS Key Management Service (KMS)
- S3 Bucket Policies
- S3 Versioning
- S3 Access Logging
- AWS Management Console

Excluded Components

- AWS CloudTrail
 - Amazon GuardDuty
 - AWS Security Hub
 - IAM Identity Center
-

Environment Information

Component	Description
Cloud Provider	AWS
Storage Service	Amazon S3
Encryption Service	AWS KMS
Logging Service	S3 Access Logging
Access Control	Bucket Policy
Data Recovery	S3 Versioning

Security Architecture

The architecture consists of an authenticated AWS user interacting with a secured S3 bucket protected by multiple security layers.

Security controls implemented include:

- Block Public Access
- AWS KMS Encryption
- Bucket Policy Enforcement
- Versioning
- Access Logging

The logging bucket receives audit records generated by Amazon S3 access requests.

Insert Screenshot:

13-AWS-S3-Security-Architecture.png

Implementation Phase 1 – Secure Bucket Creation

Objective

Create an Amazon S3 bucket with secure default configurations.

Actions Performed

- Created Amazon S3 bucket
- Selected AWS Region
- Enabled Block Public Access
- Verified bucket creation

Security Benefit

Public access restrictions reduce the likelihood of accidental data exposure caused by misconfigured permissions.

Insert Screenshots:

01-S3-Dashboard.png

02-Bucket-Created.png

03-Public-Access-Blocked.png

04-Open-Your-Bucket.png

Implementation Phase 2 – Object Versioning

Objective

Enable version control and recovery functionality.

Actions Performed

- Enabled S3 Versioning
- Uploaded test objects
- Modified and re-uploaded files
- Verified creation of multiple object versions

Security Benefits

- Recovery from accidental deletion
- Recovery from ransomware activity
- Object history preservation
- Data integrity protection

Insert Screenshots:

05-Versioning-Enabled.png

06-Test-File-Uploaded.png

12-Version-Recovery-Demo.png

Implementation Phase 3 – Data Encryption Using AWS KMS

Objective

Protect stored data using server-side encryption.

Actions Performed

- Created customer-managed KMS key
- Configured S3 Default Encryption
- Selected SSE-KMS encryption method
- Applied encryption settings

Security Benefits

Confidentiality

Protects stored information from unauthorized disclosure.

Key Management

Provides centralized encryption key administration.

Compliance

Supports cloud security and regulatory requirements.

Insert Screenshots:

07-KMS-Key-Created.png

08-S3-KMS-Encryption.png

Implementation Phase 4 – Bucket Policy Enforcement

Objective

Restrict insecure communication channels.

Actions Performed

Implemented bucket policy denying requests using non-TLS connections.

Policy Condition:

aws:SecureTransport = false

Policy Action:

DENY

Security Benefits

- HTTPS-only access

- Secure data transmission
- Reduced interception risks
- Policy-based access enforcement

Insert Screenshot:

09-Bucket-Policy.png

Implementation Phase 5 – Access Logging

Objective

Enable monitoring and auditing of bucket activity.

Actions Performed

- Created dedicated logging bucket
- Enabled Server Access Logging
- Generated bucket activity
- Verified log delivery

Security Benefits

Visibility

Provides insight into storage access activity.

Investigation Support

Supports incident response and forensic analysis.

Auditing

Maintains records of bucket access events.

Insert Screenshots:

10-Log-Bucket-Created.png

11-Access-Logs.png

Security Control Assessment

Security Control	Status
Block Public Access	Implemented
S3 Versioning	Implemented
AWS KMS Encryption	Implemented
Bucket Policy Enforcement	Implemented
Access Logging	Implemented
Recovery Validation	Implemented

Risk Analysis

Risks Before Implementation

Risk	Severity
Public Data Exposure	High
Unencrypted Storage	High
Data Loss	Medium
Insecure Communication	High
Lack of Auditing	Medium

Risks After Implementation

Risk	Residual Severity
Public Data Exposure	Low
Unencrypted Storage	Low
Data Loss	Low
Insecure Communication	Low
Lack of Auditing	Low

Key Security Findings

The implementation successfully established multiple layers of security controls within Amazon S3.

Notable improvements included:

- Elimination of public access exposure
 - Encryption of stored data using AWS KMS
 - Enforcement of secure HTTPS communication
 - Recovery capabilities through object versioning
 - Improved visibility through access logging
-

Lessons Learned

This project demonstrated how multiple AWS security controls work together to protect cloud-hosted data.

Key areas of learning included:

- Secure cloud storage architecture
 - Encryption key management
 - Access control implementation
 - Audit logging configuration
 - Data recovery mechanisms
 - AWS security best practices
-

Conclusion

The End-to-End Amazon S3 Security Implementation project successfully demonstrated the deployment of a secure cloud storage environment using AWS native security controls.

Through the implementation of public access restrictions, AWS KMS encryption, bucket policies, access logging, and versioning, the environment achieved improved confidentiality, integrity, availability, and auditability.

The project reflects practical cloud security engineering skills applicable to real-world cloud environments and demonstrates proficiency in securing Amazon S3 resources according to AWS security best practices.